

Discussion 1c Problems

Problem 1: Induction

Part (a): For all natural numbers $n > 2$, $2^n > 2n + 1$.

Base case. Let $n = 3$: $2^3 = 8 > 7 = 2(3) + 1$. ✓

Inductive hypothesis. Assume $2^n > 2n + 1$ holds for some $n = k \in \mathbb{N}$.

Inductive step. We want to show $2^{k+1} > 2(k+1) + 1 = 2k + 3$.

$$\begin{aligned} 2^{k+1} &= 2 \cdot 2^k \\ &> 2(2k + 1) \quad (\text{by inductive hypothesis}) \\ &= 4k + 2 \\ &> 2k + 3 \quad (\text{since } k > 2) \end{aligned}$$

Therefore $2^{k+1} > 2(k+1) + 1$, completing the induction. □

Part (b): For all positive integers n , $1^3 + 3^3 + 5^3 + \cdots + (2n-1)^3 = n^2(2n^2 - 1)$.

Base case. Let $n = 1$: $1^3 = 1 = 1^2(2(1)^2 - 1) = 1$. ✓

Inductive hypothesis. Assume $\sum_{j=1}^n (2j-1)^3 = n^2(2n^2 - 1)$ holds for some $n = k \in \mathbb{N}$.

Inductive step. We want to show:

$$\sum_{j=1}^{k+1} (2j-1)^3 = (k+1)^2(2(k+1)^2 - 1)$$

Starting from the left side:

$$\sum_{j=1}^{k+1} (2j-1)^3 = \sum_{j=1}^k (2j-1)^3 + (2k+1)^3$$

Applying the inductive hypothesis to the first sum:

$$= k^2(2k^2 - 1) + (2k+1)^3$$

We expand each term:

$$k^2(2k^2 - 1) = 2k^4 - k^2 \quad (\text{A})$$

$$(2k + 1)^3 = 8k^3 + 12k^2 + 6k + 1 \quad (\text{B})$$

Adding (A) and (B):

$$2k^4 + 8k^3 + 11k^2 + 6k + 1$$

We expand the target right-hand side $(k + 1)^2(2(k + 1)^2 - 1)$:

$$(k + 1)^2 = k^2 + 2k + 1$$

$$2(k + 1)^2 - 1 = 2k^2 + 4k + 1$$

$$(k^2 + 2k + 1)(2k^2 + 4k + 1) = 2k^4 + 8k^3 + 11k^2 + 6k + 1 \quad (\text{C})$$

Since (A)+(B) = (C), the inductive step holds. $\square$

Part (c): For all positive natural numbers n , $\frac{5}{4} \cdot 8^n + 3^{3n-1}$ is divisible by 19.

Base case. Let $n = 1$: $\frac{5}{4} \cdot 8 + 3^2 = 10 + 9 = 19$. $\checkmark$

Inductive hypothesis. Assume $\frac{5}{4} \cdot 8^k + 3^{3k-1} = 19m$ for some $k, m \in \mathbb{N}$.

Inductive step. We want to show $\frac{5}{4} \cdot 8^{k+1} + 3^{3(k+1)-1}$ is divisible by 19.

$$\begin{aligned} \frac{5}{4} \cdot 8^{k+1} + 3^{3k+2} &= \frac{5}{4} \cdot 8^k \cdot 8 + 3^{3k-1} \cdot 27 \\ &= 8 \left(\frac{5}{4} \cdot 8^k + 3^{3k-1} \right) + 27 \cdot 3^{3k-1} - 8 \cdot 3^{3k-1} \\ &= 8 \left(\frac{5}{4} \cdot 8^k + 3^{3k-1} \right) + 19 \cdot 3^{3k-1} \end{aligned}$$

The first term equals $8 \cdot 19m$ by the inductive hypothesis, and the second term is $19 \cdot 3^{3k-1}$. Both are divisible by 19, so their sum is divisible by 19. $\square$

Problem 2: Fibonacci Proof

Let F_i be the i th Fibonacci number defined by $F_{i+2} = F_{i+1} + F_i$, $F_0 = 0$, $F_1 = 1$. We prove:

$$\sum_{i=0}^n F_i^2 = F_n F_{n+1}$$

Base cases.

- $n = 0$: $F_0^2 = 0 = F_0 \cdot F_1 = 0 \cdot 1$. $\checkmark$
- $n = 1$: $F_0^2 + F_1^2 = 0 + 1 = 1 = F_1 \cdot F_2 = 1 \cdot 1$ (since $F_2 = F_1 + F_0 = 1$). $\checkmark$

Inductive hypothesis. Assume $\sum_{i=0}^n F_i^2 = F_n F_{n+1}$ holds for some $n = k \in \mathbb{N}$.

Inductive step. We want to show $\sum_{i=0}^{k+1} F_i^2 = F_{k+1} F_{k+2}$.

$$\begin{aligned} \sum_{i=0}^{k+1} F_i^2 &= \sum_{i=0}^k F_i^2 + F_{k+1}^2 \\ &= F_k F_{k+1} + F_{k+1}^2 \quad (\text{by inductive hypothesis}) \\ &= F_{k+1} (F_k + F_{k+1}) \\ &= F_{k+1} \cdot F_{k+2} \quad (\text{by the Fibonacci recurrence}) \end{aligned}$$

□

Problem 3: Make It Stronger

Let $a_1 = 1$ and $a_{n+1} = 3a_n^2$ for $n \geq 1$. We want to prove $a_n \leq 3^{2^n}$ for every positive integer n .

Part (a): Why the naive induction hypothesis fails.

Suppose we use the inductive hypothesis $a_k \leq 3^{2^k}$ and try to show $a_{k+1} \leq 3^{2^{k+1}}$. By definition:

$$a_{k+1} = 3a_k^2 \leq 3 \cdot \left(3^{2^k}\right)^2 = 3 \cdot 3^{2^{k+1}} = 3^{2^{k+1}+1}$$

This gives $a_{k+1} \leq 3^{2^{k+1}+1}$, but we need $a_{k+1} \leq 3^{2^{k+1}}$. The bound is off by a factor of 3 that cannot be eliminated, so the inductive hypothesis is too weak to close the proof.

Part (b): Proving the stronger statement $a_n \leq 3^{2^n-1}$.

We prove $a_n \leq 3^{2^n-1}$ by induction. Note that this implies the original claim since $2^n - 1 < 2^n$ gives $3^{2^n-1} < 3^{2^n}$.

Base case. $n = 1$: $a_1 = 1 \leq 3^{2^1-1} = 3^1 = 3$. ✓

Inductive hypothesis. Assume $a_k \leq 3^{2^k-1}$ for some $k \in \mathbb{N}$.

Inductive step. We want to show $a_{k+1} \leq 3^{2^{k+1}-1}$.

$$\begin{aligned} a_{k+1} &= 3a_k^2 \leq 3 \cdot \left(3^{2^k-1}\right)^2 \\ &= 3 \cdot 3^{2(2^k-1)} \\ &= 3^{2 \cdot 2^k - 2 + 1} \\ &= 3^{2^{k+1} - 1} \end{aligned}$$

The stronger statement holds, and therefore $a_n \leq 3^{2^n-1} \leq 3^{2^n}$ for all positive integers n . □

Problem 4: Bit String

Claim. Every positive integer n can be written as $n = c_k \cdot 2^k + c_{k-1} \cdot 2^{k-1} + \cdots + c_1 \cdot 2^1 + c_0 \cdot 2^0$ where $k \in \mathbb{N}$ and $c_i \in \{0, 1\}$.

Proof by strong induction on n .

Base case. $n = 1 = 1 \cdot 2^0$. ✓

Strong inductive hypothesis. Assume every positive integer $m < n$ has a binary representation.

Inductive step. Consider $n > 1$. We split into two cases:

Case 1: n is even. Then $n/2$ is a positive integer with $n/2 < n$. By the inductive hypothesis, $n/2$ has a binary representation $c_k \cdot 2^k + \cdots + c_0 \cdot 2^0$. Multiplying by 2 gives:

$$n = c_k \cdot 2^{k+1} + \cdots + c_0 \cdot 2^1 + 0 \cdot 2^0$$

which is a valid binary representation of n with $c_{-1} = 0$ appended.

Case 2: n is odd. Then $n - 1$ is even and $n - 1 < n$. By the inductive hypothesis, $n - 1$ has a binary representation. Since $n - 1$ is even its last bit is 0; replacing that bit with 1 gives a binary representation of n .

In both cases n has a binary representation, completing the induction. □